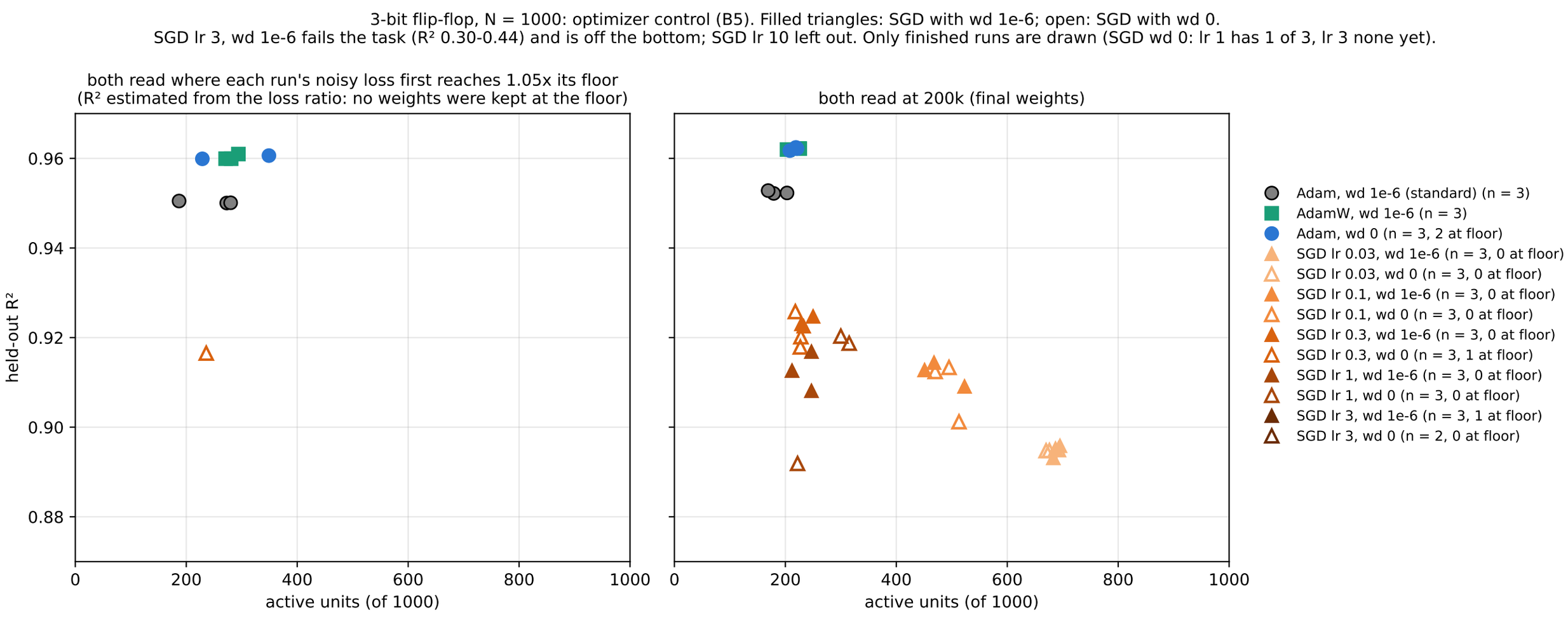
3-bit flip-flop, N = 1000: optimizer control (B5). Filled triangles: SGD with wd 1e-6; open: SGD with wd 0.
SGD lr 3, wd 1e-6 fails the task (R² 0.30-0.44) and is off the bottom; SGD lr 10 left out. Only finished runs are drawn (SGD wd 0: lr 1 has 1 of 3, lr 3 none yet).
both read where each run's noisy loss first reaches 1.05x its floor
(R² estimated from the loss ratio: no weights were kept at the floor)
both read at 200k (final weights)
held-out R²
0.96
0.94
0.92
0.90
0.88
0
200
400
600
800
1000
active units (of 1000)
Adam, wd 1e-6 (standard) (n = 3)
AdamW, wd 1e-6 (n = 3)
Adam, wd 0 (n = 3, 2 at floor)
SGD lr 0.03, wd 1e-6 (n = 3, 0 at floor)
SGD lr 0.03, wd 0 (n = 3, 0 at floor)
SGD lr 0.1, wd 1e-6 (n = 3, 0 at floor)
SGD lr 0.1, wd 0 (n = 3, 0 at floor)
SGD lr 0.3, wd 1e-6 (n = 3, 0 at floor)
SGD lr 0.3, wd 0 (n = 3, 1 at floor)
SGD lr 1, wd 1e-6 (n = 3, 0 at floor)
SGD lr 1, wd 0 (n = 3, 0 at floor)
SGD lr 3, wd 1e-6 (n = 3, 1 at floor)
SGD lr 3, wd 0 (n = 2, 0 at floor)